

DeIR-Dual: Dual-Query Instruction Rewriting for Training-Free Instruction-Following Retrieval

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Abstract

Instruction-following retrieval requires systems to adapt their ranking when query instructions change. Existing query rewriting methods (e.g., HyDE, Query2Doc, DeepRetrieval) improve raw retrieval quality through semantic enhancement but lose instruction sensitivity (p-MRR ≈ 0). We propose DeIR-Dual, which decomposes an instruction into a positive enhancement query (Q_{pos}) and a negative exclusion query (Q_{minus}), then applies a training-free decision-layer scoring mechanism that estimates background leakage in the negative channel and penalizes only residual exclusion evidence. Our experiments show that (1) inference-time score calibration derives query-specific α_q and β_q from candidate-set score distributions, avoiding labeled validation data and test-set grid search; (2) DeIR-Dual achieves the strongest average instruction sensitivity and retrieval quality among the compared FollowIR methods; (3) additional text-retrieval evaluations confirm that the decision-layer formulation transfers beyond FollowIR; and (4) we identify and analyze a *semantic entanglement* failure mode with a conservative prompting remedy.

Code: <https://github.com/...>

1 Introduction

Query rewriting has emerged as a powerful paradigm for improving dense retrieval, with methods such as HyDE [4], Query2Doc [19], and DeepRetrieval [9] demonstrating substantial gains in retrieval quality by enriching the query’s semantic coverage. Some methods even generate multiple query variants — RAG-Fusion [13] produces diverse rephrasings and fuses their retrieval results. However, all existing rewriting methods share a critical limitation: their rewritten queries are **semantically unidirectional** — they all point toward “what to retrieve” but none creates an independent channel for “what to exclude.” Even RAG-Fusion’s multiple variants are still positive rephrasings of the same intent. As a result, current rewriting remains largely **insensitive to instruction changes**: when the exclusion constraint in an instruction is modified, rankings barely move and p-MRR stays near zero [7]. Existing rewriting methods can express *where to look*, but they cannot independently express *where not to look*.

This limitation has a structural origin. Real-world retrieval instructions contain **three distinct types of signals** — background context, positive refinement, and negative exclusion [7] — yet existing rewriting pipelines only strengthen the first two. They can describe relevance more richly, but they still collapse all instruction signals into a single retrieval direction. The challenge is therefore not only to generate a negative query, but to distinguish exclusion evidence that warrants suppression from negative-channel similarity induced by topical background overlap.

Prior work attacks this problem from two other directions, but both have clear limitations. Supervised instruction-following retrievers such as Promptriever [20], InF-IR [25], and Dual-View Training [23] improve instruction following through fine-tuning on annotated or synthesized pairs, but the data cost is substantial and generalization to unseen exclusion types remains uncertain. Representation-layer negation methods such as DEO [15, 8] explicitly model positive and negative components, but resolve them through vector-space operations such as $Q_{\text{base}} - Q_{\text{neg}}$, which couples suppression with distortion of the positive intent. In both cases, instruction resolution still happens primarily at the **representation layer**.

We instead argue that instruction-following retrieval needs a **decision-layer** treatment. The failure of positive-only rewriting is not that the rewrites are weak, but that they are structurally incomplete: no collection of positive rephrasings can mark out a semantic boundary around documents targeted for exclusion. Moreover, relevant documents and “trap documents” often share topic, entities, and vocabulary, so raw negative similarity is itself confounded by background overlap. The solution is therefore twofold: create an explicit exclusion channel, and interpret that channel through a query-adaptive boundary that discounts background leakage before applying suppression.

We propose DeIR-Dual (**D**ecomposition-based **I**nstruction-**R**ewriting with **D**ual-track scoring), a training-free framework built around these two ideas. First, DeIR-Dual decomposes each instruction into a base query (Q_{base}), a positive refinement query (Q_{pos}), and a negative exclusion query (Q_{minus}), then applies a **query-adaptive exclusion boundary**: it estimates how much of the negative-channel score is expected from background relevance and penalizes only the residual excess beyond that estimate. The same residual also gates the positive reward, so documents are helped by the refinement channel only when their negative evidence remains within the safe region. Second, DeIR-Dual uses **label-free inference-time score calibration** to derive query-specific reward and penalty scales directly from the candidate-set score distributions, avoiding labeled validation data and benchmark-specific grid search.

Rather than simply producing stronger query rewrites, DeIR-Dual converts positive-only rewriting into exclusion-aware reranking without altering the retriever. On FollowIR [7], DeIR-Dual improves both instruction sensitivity and retrieval quality over query rewriting and retriever baselines. On additional negation-sensitive and general-domain benchmarks, the same decision-layer formulation transfers without per-dataset supervision. Finally, diagnostic analysis identifies **semantic entanglement** — heavy overlap between the base and exclusion channels — as the key failure mode that limits training-free exclusion reasoning.

In summary, we make two technical contributions and one empirical observation: **(i)** a query-adaptive exclusion boundary that converts triple-query decomposition into selective decision-layer suppression; **(ii)** a label-free inference-time calibration scheme that aligns the boundary and reward scales to each query’s candidate-set statistics; and **(iii)** an empirical characterization of semantic entanglement as the main boundary-failure regime.

2 Related Work

2.1 LLM-Based Query Rewriting and Semantic Enhancement

LLM-based query rewriting improves retrieval by making the user’s information need more explicit before matching. HyDE [4] retrieves with hypothetical documents, Query2Doc [19] enriches queries with pseudo-documents, and RAG-Fusion [13] combines multiple generated query variants. More recent rewriting systems learn the rewriting policy or optimize retrieval-oriented generation, including RAG-QR [10] and DeepRetrieval [9]. These methods primarily improve recall or semantic coverage, but they do not explicitly model exclusion as a separate decision signal. Even when multiple rewrites are generated, the variants usually remain positive descriptions of relevance rather than independent evidence for what should be suppressed. Our method may use query decomposition to obtain positive and negative channels, but moves beyond rewriting by introducing an explicit exclusion-aware decision layer.

2.2 Instruction-Following Retrieval with Exclusion Constraints

Instruction-following retrieval extends dense retrieval from topical matching to instruction-conditioned matching. Instruction-tuned retrievers such as INSTRUCTOR [14] and TART [1] train embedding models with task descriptions, while larger instruction-aware retrievers such as E5-Mistral [17], BGE/C-Pack [21], GritLM [11], Promptriever [20], and InF-IR [25] improve instruction sensitivity through instruction-labeled or synthetic supervision. FollowIR [7] further formalizes per-query instruction following through original/changed query pairs and metrics such as p-MRR, while Dual-View Training [23] synthesizes polarity-reversal data to improve sensitivity to instruction changes. These works show that instruction-following retrieval can be improved through training, but training-centric approaches require retriever updates and additional instruction data, and they must balance instruction sensitivity against broad retrieval generalization.

Exclusion constraints are a particularly challenging form of instruction following. In exclusion queries, the hardest negatives are not irrelevant documents but topically relevant documents that violate the constraint. Such documents may match the affirmative intent and share substantial background with the query, yet violate the user’s “not”, “without”, or “exclude” condition. This makes positive query enhancement alone less suited to the problem: it can strengthen topical relevance without identifying which relevant-looking documents should be suppressed.

One line of work addresses negative constraints and complex logical queries by making constraint structure explicit. Neural-symbolic retrieval methods such as NS-IR [22] convert natural-language queries and documents into first-order logic forms and enforce logical consistency through reranking objectives such as logic alignment and connective constraints. OrLog [6] instead decomposes complex queries into atomic predicates, estimates predicate-level plausibility with LLMs, and applies probabilistic reasoning to infer query satisfaction. These routes offer interpretable ways to represent logical constraints, but they introduce a different dependency: natural-language retrieval must be converted into logical or predicate-level representations, or coupled with symbolic/probabilistic reasoning objectives. Our method stays in the dense score space and does not require a logical parser, symbolic inference module, or probabilistic reasoning engine.

A closer line of work avoids retriever training by decomposing instructions into positive and negative components and modifying query representations at inference time. DEO-style methods [15, 16, 8] learn or construct positive and negative instruction components, with training-free variants operating at the representation layer through query embedding modification or optimization. Representation-layer methods change the query vector globally, which can make it difficult to distinguish true exclusion evidence from background overlap at the document level. In contrast, DeIR-Dual keeps the retriever fixed, avoids symbolic query conversion, and resolves exclusion at the decision layer. By estimating background leakage and calibrating reward/penalty scales from the current unlabeled candidate set, it targets constraint satisfaction without retriever fine-tuning or global query-vector manipulation.

2.3 Query-Local Score Calibration and Test-Time Adaptation

The calibration component of DeIR-Dual is related to post-hoc calibration and local similarity correction. Platt scaling and temperature scaling transform model outputs after training while keeping model parameters fixed [12, 5], but are typically estimated from labeled validation data for confidence calibration. Local scaling methods such as CSLS [2] show a complementary principle: raw similarities can contain local bias, and neighborhood statistics can improve comparability across candidates. Our setting differs from both lines because it requires label-free calibration of ranking scores for each query.

Information retrieval also provides a direct precedent for using unlabeled query-local evidence. Query performance prediction estimates the expected reliability of a ranking without relevance judgments, and recent studies show that neural rankers exhibit query-specific behaviors that are difficult to capture with traditional predictors [3]. Pseudo-relevance feedback similarly uses the top-ranked set as local evidence, including in dense retrieval settings where feedback representations are extracted from first-pass results [18]. DeIR-Dual adopts the same broad test-time principle but uses the candidate set differently: it does not predict query difficulty or expand the query, but estimates the score scales of the base, positive, and negative channels.

Recent inference-only calibration work further shows that unlabeled test batches can be used to estimate and correct score biases without supervised validation labels [24]. In contrast, DeIR-Dual does not tune prompts, update model parameters, or learn global calibration constants. It keeps the retriever fixed and uses candidate-set statistics only to calibrate decision-layer reward and penalty scales.

3 Method: DeIR-Dual

3.1 Problem Formulation

Dense retrievers typically encode a query–instruction pair into a single vector and rank documents by cosine similarity. In instruction-following retrieval, however, the instruction may contain signals with different roles:

1. **Background context:** information that frames the user’s information need and helps interpret the instruction.
2. **Positive refinement:** affirmative constraints that specify what should be emphasized or made more specific.
3. **Negative exclusion:** constraints that specify what should be suppressed or treated as irrelevant.

When these signals are collapsed into one embedding, the resulting score provides a single relevance direction. This score can be effective for topical matching, but it does not expose separate controls for contextual relevance, positive refinement, and negative exclusion. The issue is reflected in changed-instruction evaluations such as FollowIR [7], where a system must respond to fine-grained query-level constraints rather than only retrieve topically related documents.

We formulate the problem as inference-time decision-layer reranking with a frozen retriever. Given a query and instruction, the goal is to construct separate scoring channels for contextual relevance, positive refinement, and negative exclusion without updating the encoder. The negative channel is then converted into a query-adaptive exclusion boundary rather than used as a raw penalty.

3.2 Triple-Query Decomposition

DeIR-Dual constructs three query channels that provide the decision layer with separate evidence for context, affirmative refinement, and exclusion:

1. Q_{base} (**Base Query**) contains the original query and the full instruction. It serves as the contextual relevance anchor that preserves the broader information need.
2. Q_{pos} (**Positive Refinement**) extracts the affirmative constraints that refine what should be retrieved. This channel emphasizes positive requirements, while the base channel retains the broader context.
3. Q_{minus} (**Negative Exclusion**) extracts the excluded content and defines the negative channel used for boundary estimation.

The three queries are independently encoded by the same dense retriever $f(\cdot)$, producing three query embeddings $\mathbf{h}_{\text{base}} = f(Q_{\text{base}})$, $\mathbf{h}_{\text{pos}} = f(Q_{\text{pos}})$, $\mathbf{h}_{\text{neg}} = f(Q_{\text{minus}})$. Document embeddings $\mathbf{v}_d = f(d)$ are pre-computed once and reused.

The base channel is not used to enforce exclusion directly; it provides the contextual relevance anchor against which the positive and negative channels are calibrated. This role is important because a positive/negative decomposition alone may remove contextual information that determines how refinement and exclusion should be interpreted. We therefore keep the full instruction as a context anchor and use the positive and negative queries as targeted decision-layer signals.

In practice, the decomposer is prompted to produce short retrieval-style queries and to place affirmative and excluded constraints in separate fields. Prompts and examples are provided in the appendix.

3.3 Query-Adaptive Exclusion Boundary

Given the three query encodings, we compute document-level cosine similarities

$$S_{\text{base}}(d) = \cos(\mathbf{h}_{\text{base}}, \mathbf{v}_d), \quad S_{\text{pos}}(d) = \cos(\mathbf{h}_{\text{pos}}, \mathbf{v}_d), \quad S_{\text{neg}}(d) = \cos(\mathbf{h}_{\text{neg}}, \mathbf{v}_d). \quad (1)$$

The base and positive channels provide topical relevance and instructional refinement. The negative channel is more delicate: a high $S_{\text{neg}}(d)$ can indicate a true constraint violation, but it can also arise because relevant and excluded documents share entities, topic, or background context. DeIR-Dual therefore treats exclusion as a **residual boundary estimation** problem. Negative-channel similarity is decomposed into background leakage and residual exclusion evidence. We estimate the background component from the current query’s candidate distribution, subtract it from the negative-channel score, and penalize residual evidence only as it approaches and exceeds a robust query-specific boundary.

Background Leakage Estimation

Our treatment follows the same broad intuition as local scaling methods for nearest-neighbor retrieval: raw similarity can contain a query-local bias that should be removed before comparison [2]. In our setting, this bias is background-induced similarity in the negative channel. We use a simple query-local linear leakage model to estimate the part of the negative-channel score that is expected from background relevance. Let μ_b, σ_b be the mean and standard deviation of S_{base} over the candidate set, and let μ_n, σ_n be the corresponding statistics for S_{neg} . We first standardize the base score,

$$z_{\text{base}}(d) = \frac{S_{\text{base}}(d) - \mu_b}{\sigma_b}, \quad (2)$$

and measure the query-level overlap between the base and exclusion channels as

$$c_q = \cos(\mathbf{h}_{\text{base}}, \mathbf{h}_{\text{neg}}). \quad (3)$$

The expected negative-channel score induced by background relevance is then

$$\hat{S}_{\text{neg}}^{\text{bg}}(d) = \mu_n + \sigma_n c_q z_{\text{base}}(d). \quad (4)$$

This estimate transfers the standardized base-channel variation into the negative-channel scale, modulated by the semantic overlap between the base and exclusion queries. It is not intended as a generative model of relevance; it is a query-local baseline for the negative score expected from background overlap.

Residual Exclusion Boundary

We define the residual negative score as the excess negative-channel evidence beyond the background estimate:

$$R_{\text{neg}}(d) = S_{\text{neg}}(d) - \hat{S}_{\text{neg}}^{\text{bg}}(d). \quad (5)$$

Positive residuals indicate that the document matches the exclusion query more strongly than expected from background relevance alone. We define the query-adaptive exclusion boundary with the median and median absolute deviation of the residual scores:

$$\tau_q = \text{median}(\mathcal{R}_q) + \lambda \text{MAD}(\mathcal{R}_q), \quad \mathcal{R}_q = \{R_{\text{neg}}(d) : d \in \mathcal{C}_q\}, \quad (6)$$

where $\mathcal{C}_q$ denotes the retrieved candidate set. Residuals below τ_q are treated as background-consistent, while residuals above τ_q provide stronger evidence for exclusion. We use Softplus as a smooth approximation to a hinge penalty:

$$\text{penalty}(d) = \alpha_q \text{Softplus}(R_{\text{neg}}(d) - \tau_q). \quad (7)$$

The penalty grows smoothly as residual exclusion evidence approaches and exceeds the boundary, while the residual construction prevents the negative channel from penalizing documents merely because they share topical background with the exclusion query. The scale α_q is derived by score calibration in §3.4.

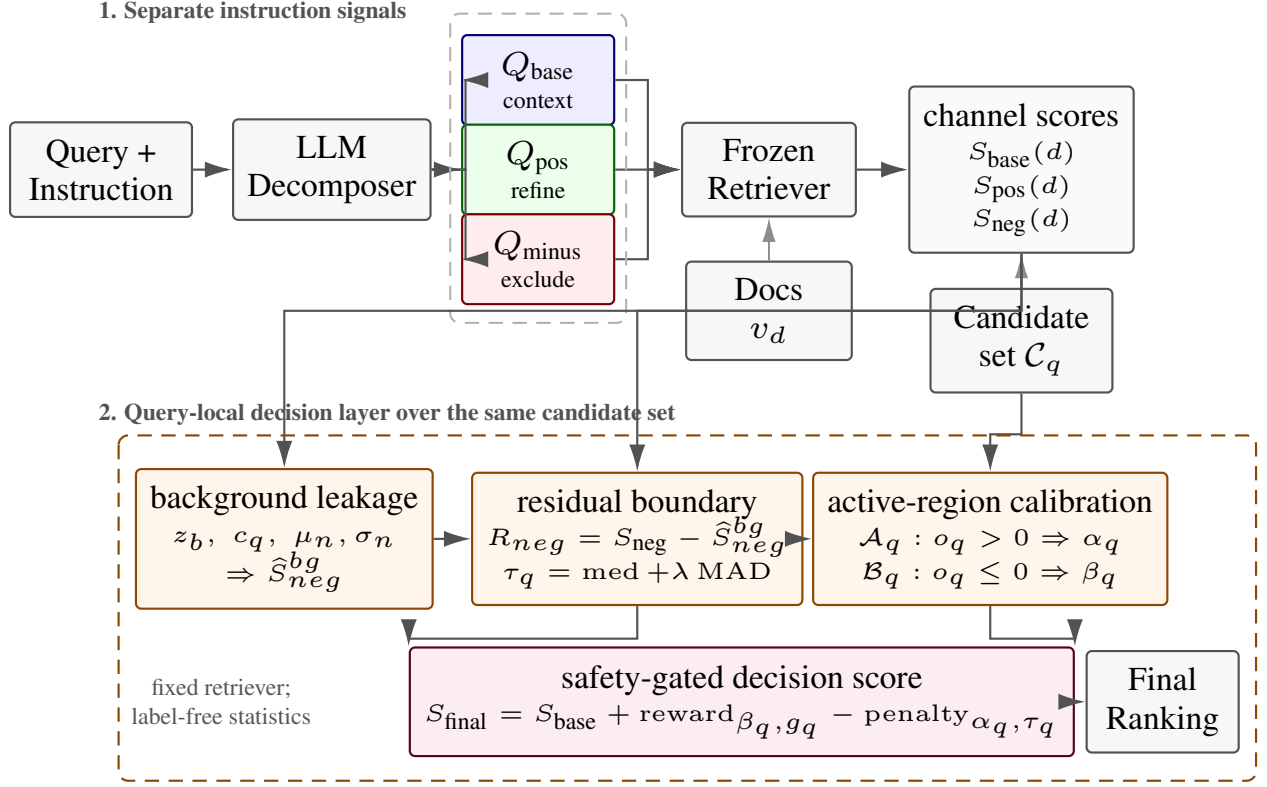


Figure 1: Overview of the current DeIR-Dual formulation. The method first separates the instruction into context, positive-refinement, and negative-exclusion channels and scores the same candidate set with a frozen retriever. The decision layer then estimates background-induced negative similarity, converts the negative score into a residual boundary signal, derives α_q and β_q from active and safe candidate regions, and composes the final safety-gated reranking score in Eq. 9.

Safety-Gated Positive Reward

The positive channel can improve recall for documents satisfying the instruction, but a raw positive reward can also amplify documents that violate the exclusion constraint. We therefore gate the positive reward with respect to the same residual boundary:

$$g_q(d) = 1 - \sigma \left(\kappa \frac{R_{\text{neg}}(d) - \tau_q}{\text{MAD}(\mathcal{R}_q)} \right). \quad (8)$$

The gate largely preserves the positive reward for documents whose residuals fall below the boundary and progressively suppresses it as residual exclusion evidence crosses the boundary. The fixed shape parameter κ controls the sharpness of this transition.

Final score. The final decision-layer score combines base relevance, safety-gated positive refinement, and residual exclusion penalty:

$$S_{\text{final}}(d) = S_{\text{base}}(d) + \beta_q S_{\text{pos}}(d) g_q(d) - \alpha_q \text{Softplus}(R_{\text{neg}}(d) - \tau_q). \quad (9)$$

This nonlinear score composition implements a query-adaptive exclusion boundary at ranking time without modifying the dense retriever. The parameters α_q and β_q are derived from the score distributions of the current query, while λ and κ are fixed shape parameters controlling boundary offset and gate sharpness.

3.4 Inference-Time Score Calibration

The scoring formula in Eq. 9 contains two query-level calibration factors: α_q for the residual exclusion penalty and β_q for the positive refinement reward. Rather than grid-searching these values on labeled queries, we derive them at reranking time from the score distributions of the current candidate set. This converts parameter selection into query-conditioned score calibration: the correction terms are placed on a magnitude comparable to the base retriever without

changing the retriever or using relevance labels. The goal is not to fit a benchmark-specific optimum, but to make the exclusion boundary and positive reward portable across retrievers with different score scales.

Candidate-Set Score Calibration

The base, positive, and negative channels are produced by the same dense retriever but have different score distributions because they encode different query intents. We therefore treat α_q and β_q as post-hoc scale factors for decision-layer reranking. Unlike validation-set calibration methods such as Platt scaling and temperature scaling [12, 5], our calibration is label-free, query-local, and operates on ranking scores. The only inputs are the channel scores over the current candidate set $\mathcal{C}_q$.

Let the residual excess over the boundary be

$$o_q(d) = R_{\text{neg}}(d) - \tau_q, \quad (10)$$

and define the residual-exceeding subset as

$$\mathcal{A}_q = \{d \in \mathcal{C}_q : o_q(d) > 0\}. \quad (11)$$

These are the documents whose negative residual exceeds the robust boundary and therefore carries residual exclusion evidence. We write the corresponding penalty magnitude as

$$P_q(d) = \text{Softplus}(o_q(d)). \quad (12)$$

Let $E_b(d)$ and $E_{\text{pos}}(d)$ denote nonnegative evidence magnitudes for the base and positive-refinement channels. These magnitudes are used only for scale estimation. We derive α_q and β_q by active-region scale matching: each correction term is scaled so that its average magnitude on the region where it is active is comparable to the average base evidence on that same region. For the penalty-active region, this gives

$$\alpha_q = \frac{\text{Avg}_{d \in \mathcal{A}_q} E_b(d)}{\text{Avg}_{d \in \mathcal{A}_q} P_q(d)}. \quad (13)$$

For the positive channel, we use the complementary safe subset

$$\mathcal{B}_q = \{d \in \mathcal{C}_q : o_q(d) \leq 0\}. \quad (14)$$

The reward coefficient is derived after applying the safety gate, since the positive reward is reduced when a document has residual negative evidence. We define the gated positive-refinement evidence as

$$E_q^+(d) = g_q(d) E_{\text{pos}}(d). \quad (15)$$

The same scale-matching principle calibrates the reward-active region: the average gated positive-refinement evidence in $\mathcal{B}_q$ is matched to the average base evidence on $\mathcal{B}_q$:

$$\beta_q = \frac{\text{Avg}_{d \in \mathcal{B}_q} E_b(d)}{\text{Avg}_{d \in \mathcal{B}_q} E_q^+(d)}. \quad (16)$$

Degenerate cases are handled by fixed label-free defaults.

Fixed Shape Parameters

The main scale parameters, α_q and β_q , are derived from the current query’s score statistics. By contrast, λ and κ are fixed shape constants: λ controls how far the boundary is placed above the residual median, and κ controls the sharpness of the gate around that boundary. They are fixed across queries and retrievers, and sensitivity analysis examines their effect.

Label-Free Inference-Time Derivation

A central risk in inference-time calibration is test-set leakage: parameters must not be selected using relevance labels, evaluation metrics, or benchmark-level grid search. Our derivation avoids this leakage because α_q and β_q are computed only from the unlabeled channel scores in $\mathcal{C}_q$. The retrieval labels are never used when computing Eqs. 13–16. The remaining constants λ and κ define the boundary offset and gate sharpness. They are fixed across queries and analyzed separately in ablation and sensitivity studies. This separation keeps the method portable across dense retrievers: query-level calibration adapts to the score scale of the current retriever, while global shape constants are not re-selected for individual test queries.

	Model	Robust04		News21		Core17		Average	
		MAP	p-MRR	nDCG	p-MRR	MAP	p-MRR	Score	p-MRR
Cross Encoders	MonoT5-3B	27.3	+4.0	16.5	+1.8	18.2	+1.8	20.7	+2.5
	Mistral-7B-instruct	23.2	+12.6	27.2	+4.8	19.7	+13.0	23.4	+10.1
	FollowIR-7B	24.8	+13.7	29.6	+6.3	20.0	+16.5	24.8	+12.2
Bi Encoders	BM25	12.1	-3.1	19.3	-2.1	8.1	-1.1	13.2	-2.1
	TART-Contriever	14.3	-9.0	21.8	-3.0	13.3	-3.0	16.5	-5.0
	Instructor XL	19.7	-8.1	26.1	-0.9	16.8	0.7	20.9	-2.8
	E5-Mistral	23.1	-9.6	27.8	-0.9	18.3	+0.1	23.1	-3.5
	Google Gecko	23.3	-2.4	29.5	+3.9	23.2	+5.4	25.3	+2.3
	RepLLaMA	24.0	-8.9	24.5	-1.8	20.6	+1.3	23.0	-3.1
	Promptriever	28.3	+11.7	28.5	+6.4	21.6	+15.4	26.1	+11.2
	DeIR-Dual	28.6	+8.3	32.3	+18.7	26.0	+17.6	29.0	+14.9

Table 1: Performance of existing retrieval methods and DeIR-Dual on the FollowIR benchmark [7]. All metrics are percentages ($\times 100$). DeIR-Dual achieves the highest News21 nDCG, Core17 MAP, average score, and average p-MRR without any fine-tuning.

4 Experiments

4.1 Evaluation Protocol

We evaluate DeIR-Dual as an inference-time reranking method with frozen retrievers. The main experiments measure whether the same decision-layer scorer improves instruction-following retrieval quality across datasets and retrievers without retriever fine-tuning or labeled calibration. On FollowIR, we report both instruction sensitivity (p-MRR) and task quality (MAP/nDCG and the aggregate Score). On transfer benchmarks, we lead with the task-quality metric appropriate to that benchmark and use p-MRR-style evidence only when the benchmark explicitly evaluates instruction changes.

The experimental section first presents the main comparison against retrieval and query-rewriting baselines, followed by cross-retriever results and negation-sensitive transfer. We then use two compact ablation tables to isolate the method’s two central mechanisms: the residual exclusion boundary with safety-gated reward, and query-local α_q, β_q calibration. Sensitivity to λ, κ , and candidate set size, together with runtime overhead and additional standard-retrieval controls, is reported in the appendix.

We use FollowIR [7] as the primary instruction-following benchmark. Table 1 summarizes the reported performance of representative cross encoders (MonoT5-3B, Mistral-7B-instruct, FollowIR-7B) and bi-encoder dense models (BM25, TART-Contriever, Instructor XL, E5-Mistral, Google Gecko, RepLLaMA, Promptriever) on its three TREC collections. Most off-the-shelf retrievers exhibit near-zero or negative p-MRR, indicating limited sensitivity to instruction changes. Promptriever obtains the strongest baseline score after supervised instruction tuning, whereas DeIR-Dual keeps the retriever fixed and applies only inference-time reranking.

4.2 Main Results

Instruction Following on FollowIR

Table 2 reports results on FollowIR. Among the inference-time rewriting and reranking baselines, DeIR-Dual obtains the best mean p-MRR (14.9) and aggregate Score (29.0). Compared with the strongest rewriting baseline, INF-X Full, DeIR-Dual improves p-MRR by 11.5 points and Score by 2.0 points. It also obtains the strongest aggregate retrieval quality among these methods, with the highest Core17 MAP and News21 nDCG. This indicates that the gain in instruction sensitivity does not come at the cost of standard retrieval quality. Table 1 further provides frozen retriever and supervised instruction-following reference baselines for context: compared with the supervised Promptriever reference, DeIR-Dual improves average Score by 2.9 points and average p-MRR by 3.7 points while keeping the retriever fixed. These results show that explicit exclusion-aware reranking is more effective than positive-only query enhancement for changed-instruction retrieval.

Method	Core17		Robust04		News21		Average	
	MAP	p-MRR	MAP	p-MRR	nDCG	p-MRR	Score	p-MRR
DeepRetrieval	21.5	+6.6	25.9	-5.3	23.9	-2.2	23.8	-0.3
HyDE	23.6	+6.5	24.9	-2.9	21.4	+0.7	23.4	+1.4
Query2Doc	25.9	+8.0	27.9	-7.5	24.9	-3.8	26.2	-1.1
RAG-Fusion	21.9	+5.4	21.1	-8.1	21.0	+1.8	21.3	-0.3
RAG-QR	22.3	+2.3	27.6	-10.3	22.9	-2.1	24.3	-3.4
ConvSearch-R1	22.3	+3.0	25.9	-2.5	22.0	+0.2	23.4	+0.2
INF-X Full	27.0	+7.2	32.1	-1.4	22.3	+4.4	27.0	+3.4
BGE-Reasoner-Rewriter	21.5	+0.7	–	–	20.5	+3.4	–	–
ReFeed-8B	18.3	-6.6	23.3	-1.2	22.4	-0.4	21.2	-2.7
TongSearch-QR	21.7	+3.3	24.5	-4.6	22.6	+0.2	22.7	-0.4
DeIR-Dual	26.0	+17.6	28.6	+8.3	32.3	+18.7	29.0	+14.9

Table 2: Comparison of DeIR-Dual with ten query rewriting methods on the FollowIR benchmark. DeIR-Dual achieves the highest mean p-MRR (14.9) and Score (29.0), improving over the strongest query rewriting baseline (INF-X Full) by 11.5 p-MRR points and 2.0 Score points. All metrics are percentages ($\times 100$).

Retriever	Base Retriever		Strong Inference-Time Baseline		DeIR-Dual	
	Score	p-MRR	Score	p-MRR	Score	p-MRR
E5-Mistral	TBD	TBD	TBD	TBD	TBD	TBD
BGE	TBD	TBD	TBD	TBD	TBD	TBD
GritLM	TBD	TBD	TBD	TBD	TBD	TBD
ReplLaMA	TBD	TBD	TBD	TBD	TBD	TBD

Table 3: Cross-retriever main results on FollowIR. The table reports the average task-quality Score and instruction-sensitivity p-MRR for each frozen retriever. The same DeIR-Dual scoring rule and fixed λ, κ are used across retrievers; only query-local α_q, β_q are derived from the current candidate set.

Portability Across Frozen Retrievers

Table 3 evaluates DeIR-Dual across frozen retrievers with different score scales and embedding spaces. For each retriever, reranking is performed on the same top- K candidate set returned by the base model. The same shape constants are used for all retrievers, while α_q and β_q are derived from each query’s candidate scores. This setting tests whether query-local calibration is tied to a single encoder or can be applied as a portable decision-layer reranker over different frozen retrievers.

Transfer to Negation-Sensitive Benchmarks

Table 4 evaluates transfer to NegConstraint, a text-retrieval benchmark where exclusion is expressed directly as a retrieval constraint. Following the NegConstraint protocol [22], we report the total MAP and nDCG@10 in the main text and defer formulation-level results to Appendix B. We also include DEO [8], a close training-free negation-aware retrieval baseline. The method is transferred without dataset-specific retraining or coefficient search: the same scoring mechanism and fixed shape constants are used, while α_q and β_q are derived from unlabeled candidate-set scores. This setting tests whether the residual boundary mechanism generalizes beyond FollowIR’s changed-instruction protocol and whether it improves over positive-only refinement and raw negative penalties.

4.3 Ablation Study

The main text keeps the ablation study deliberately compact. The first table tests the residual-boundary mechanism, and the second table tests query-local score calibration. Other implementation and robustness checks are deferred to Appendix A so that each main-text table answers a single reviewer question.

Method	MAP	nDCG@10
BM25	31.4	33.7
Contriever	31.8	35.7
BGE	36.3	40.8
HyDE	47.8	53.1
InteR	52.3	54.5
NS-IR (GPT-4o)	53.3	56.5
BGE-S	67.0	73.7
BGE-S w/ DEO	73.0	78.0
BGE-L	63.0	71.4
BGE-L w/ DEO	73.3	78.8
BGE-M3	63.7	72.5
BGE-M3 w/ DEO	73.8	79.5
Base retriever	TBD	TBD
+ Positive channel	TBD	TBD
+ Raw negative penalty	TBD	TBD
DeIR-Dual	TBD	TBD

Table 4: Transfer to NegConstraint [22]. The main text reports total MAP and nDCG@10; formulation-level results are deferred to Appendix B. DEO results are reported in its arXiv preprint [8]; BGE-S and BGE-L denote BGE-small-en-v1.5 and BGE-large-en-v1.5. All scores are percentages ($\times 100$).

Variant	Core17		Robust04		News21		Average	
	MAP	p-MRR	MAP	p-MRR	nDCG	p-MRR	Score	p-MRR
Base retriever	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
+ Positive channel	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
+ Raw negative penalty	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
+ Residual boundary	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Full DeIR-Dual	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Table 5: Core mechanism ablation on FollowIR. All variants use the same frozen encoder and candidate pool. The rows isolate positive refinement, direct raw negative-channel penalization, residual-boundary penalization, and the full scorer with safety-gated query-local calibration. All metrics are percentages ($\times 100$).

Scale Selection	Core17		Robust04		News21		Average	
	MAP	p-MRR	MAP	p-MRR	nDCG	p-MRR	Score	p-MRR
Fixed α, β	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Global calibration	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Query-local calibration	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Grid-search oracle	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Table 6: Calibration ablation on FollowIR. All rows use the same residual boundary and safety gate; only the selection of α and β changes. The oracle row is used only as an analysis upper bound for tuned alternatives, not as a deployable method. All metrics are percentages ($\times 100$).

Core Mechanism: From Positive Rewriting to Residual Boundary

Table 5 attributes the first technical contribution by comparing positive-only refinement, raw negative-channel penalization, residual-boundary penalization, and the full safety-gated calibrated scorer under the same retrieval setting.

Query-Local Calibration

Table 6 tests whether the gains can be explained by ordinary parameter tuning. Improvements from query-local calibration over fixed and global coefficients support the active-region scale-matching principle, while the oracle row quantifies remaining headroom under benchmark-level tuning.

Variant	Change from Full DeIR-Dual	Avg Score	Avg p-MRR	Runtime
Full DeIR-Dual	None	TBD	TBD	TBD
w/o safety gate	Remove $g_q(d)$ from positive reward	TBD	TBD	TBD
α_q only	Query-local penalty scale, fixed reward scale	TBD	TBD	TBD
β_q only	Query-local reward scale, fixed penalty scale	TBD	TBD	TBD
Mean/std boundary	Replace median/MAD boundary with mean/std	TBD	TBD	TBD
Raw s_n boundary	Estimate boundary on raw negative similarity	TBD	TBD	TBD

Table 7: Diagnostic ablations for secondary design choices. The table keeps the same frozen retriever and candidate pool as Table 5; missing values are placeholders to be filled after running the corresponding controlled variants.

Analysis	Setting	Avg Score	Avg p-MRR	Latency	Notes
λ sensitivity	$\lambda \in \{\text{TBD}\}$	TBD	TBD	TBD	Boundary offset
κ sensitivity	$\kappa \in \{\text{TBD}\}$	TBD	TBD	TBD	Gate sharpness
Candidate set size	$K \in \{\text{TBD}\}$	TBD	TBD	TBD	Reranking depth
Standard retrieval control	BEIR/TREC subset	TBD	–	TBD	No instruction changes
Failure subset	High semantic-overlap queries	TBD	TBD	TBD	Entanglement cases

Table 8: Sensitivity, efficiency, and robustness result template. These analyses are intended to accompany the main ablations without changing the deployed scoring rule; all missing values should be filled from matched evaluation runs.

We fix λ and κ across all retrievers and datasets; sensitivity results in the appendix verify that performance is stable over a broad range. Additional analyses cover candidate-set size sensitivity, α_q -only and β_q -only variants, mean/std versus median/MAD boundary estimation, runtime overhead, robustness on standard retrieval benchmarks, and semantic-entanglement failure cases.

5 Conclusion and Limitations

This paper presented DeIR-Dual, a frozen-retriever decision-layer reranking method for text retrieval under exclusion-sensitive instructions. The central idea is to avoid treating negative-query similarity as uniformly harmful. DeIR-Dual first uses triple-query decomposition to separate contextual relevance, positive refinement, and exclusion evidence; it then estimates a query-adaptive residual exclusion boundary over the candidate set and applies label-free query-local score calibration to place residual penalties and gated positive rewards on comparable scales. The current FollowIR results show improvements in instruction sensitivity and aggregate retrieval quality over inference-time rewriting and reranking baselines, while the cross-retriever, transfer, ablation, and appendix sensitivity analyses are designed to test whether these gains persist across frozen retrievers, datasets, mechanism variants, and fixed shape constants once their remaining entries are completed.

The scope of DeIR-Dual is bounded by several limitations. It depends on LLM-generated decompositions, so ambiguous or underspecified instructions can propagate errors into the positive and negative channels. High-overlap cases, where desired and excluded semantics are tightly entangled, remain difficult because residual exclusion evidence can be hard to distinguish from background relevance. The fixed shape constants λ and κ are intended to improve portability, but their stability still requires the planned sensitivity validation. Finally, because DeIR-Dual reranks an existing candidate set rather than retrieving new documents, its effectiveness depends on the base retriever placing relevant documents in $\mathcal{C}_q$; missed candidates cannot be recovered by the decision-layer scorer.

A Additional Ablations and Sensitivity

Table 7 separates smaller design choices that should not be conflated with the two main ablations. These rows are used to check whether the proposed robust boundary, safety gate, and two-sided calibration each contribute beyond simpler replacements.

Table 8 reports the planned robustness and efficiency checks. The λ and κ sweeps verify that the fixed shape constants are not serving as hidden test-set tuning, while candidate-set and runtime analyses quantify the inference-time cost of candidate-set score calibration.

Method	$A - a$		$(A - a) \cup B$		$(A - a) \cup (B - b)$		Total	
	MAP	nDCG@10	MAP	nDCG@10	MAP	nDCG@10	MAP	nDCG@10
BM25	32.1	34.6	31.2	34.7	29.3	31.5	31.4	33.7
Contriever	34.8	36.6	32.1	33.3	30.9	32.7	31.8	35.7
BGE	37.9	40.5	34.8	36.8	33.7	34.8	36.3	40.8
HyDE	50.7	55.3	48.6	51.5	45.7	50.6	47.8	53.1
InteR	52.6	55.8	50.3	48.7	51.5	49.3	52.3	54.5
NS-IR (GPT-4o)	54.7	57.9	53.3	54.2	51.7	53.7	53.3	56.5
BGE-S	—	—	—	—	—	—	67.0	73.7
BGE-S w/ DEO	—	—	—	—	—	—	73.0	78.0
BGE-L	—	—	—	—	—	—	63.0	71.4
BGE-L w/ DEO	—	—	—	—	—	—	73.3	78.8
BGE-M3	—	—	—	—	—	—	63.7	72.5
BGE-M3 w/ DEO	—	—	—	—	—	—	73.8	79.5
Base retriever	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
+ Positive channel	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
+ Raw negative penalty	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
DeIR-Dual	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Table 9: Formulation-level NegConstraint results. ACL reference results are taken from the NegConstraint paper [22]. DEO results are total-only numbers reported in its arXiv preprint [8]. All scores are percentages ($\times 100$).

B Additional NegConstraint Results

Table 9 provides the formulation-level NegConstraint results corresponding to the total scores reported in Table 4. DEO reports only total scores in the referenced preprint, so its formulation-level entries are left blank.

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